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Exercise No: 1

Date: 26/08/2025

CREATING COLOUR COMPOSITES

Objective: To download Landsat 8/9 Collection 2 Level 2 satellite images from USGS Earth Explorer and generate true color and false color composites using QGIS.

Software Required:

- **QGIS (latest version installed)**
- **Internet access to download satellite data from USGS Earth Explorer**

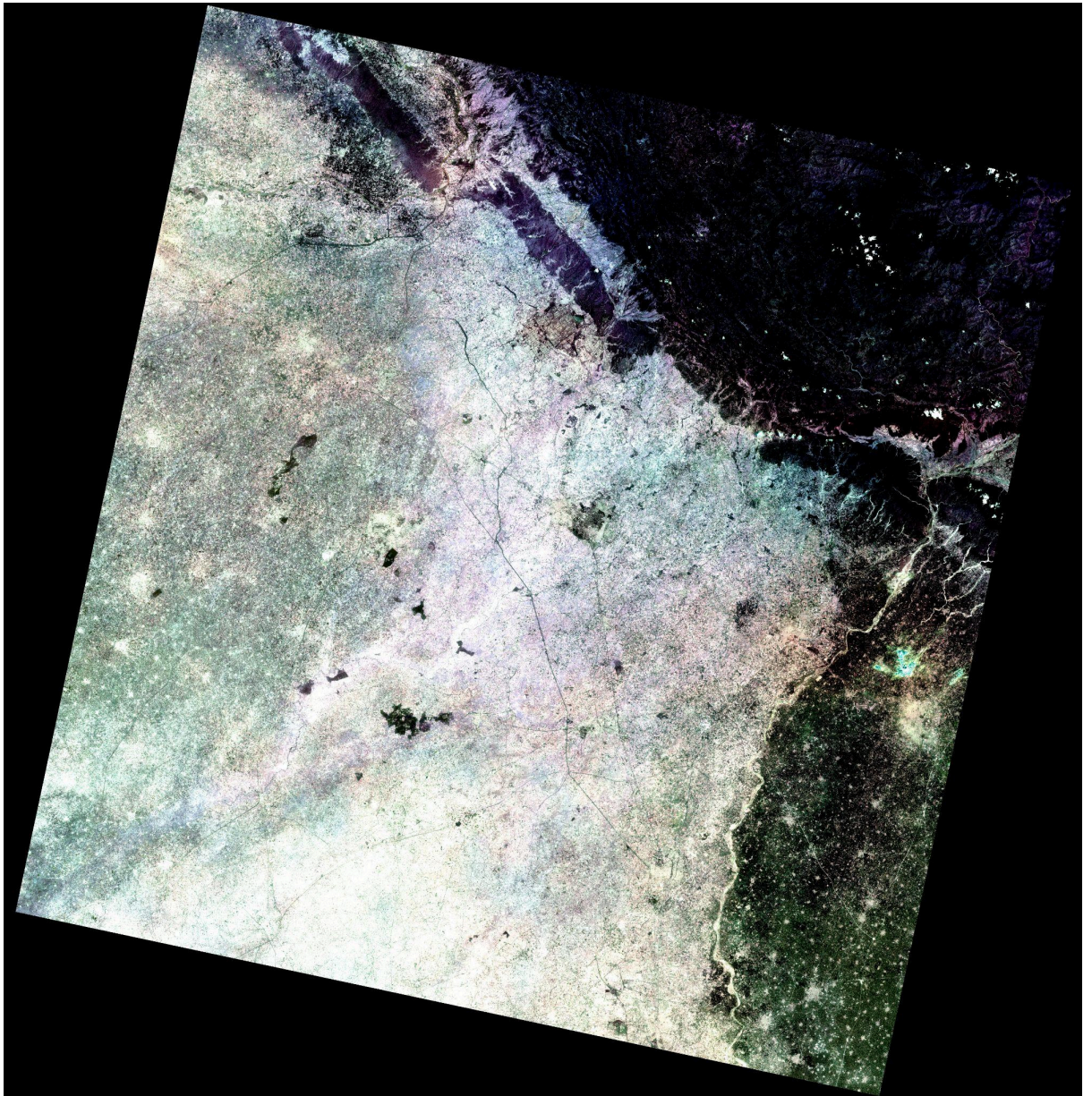
Procedure:

1. Data Download
 - Open USGS Earth Explorer.
 - Select any date range with Landsat Collection 2 Level 2 data.
 - Apply filter: Cloud cover between 0–5%.
 - Download required bands (Band 2, Band 3, Band 4, Band 5).
2. Import Data in QGIS
 - Open QGIS → Data Source Manager → Raster.
 - Browse and load the four bands (B2, B3, B4, B5).
3. Create Virtual Raster (to combine bands)
 - Go to *Raster tab* → *Miscellaneous* → *Build Virtual Raster*.
 - Select the four input bands.
 - Save the virtual raster file.
4. True Color Composite
 - Right-click the virtual raster → Properties → Symbology.
 - Assign Red = Band 3, Green = Band 2, Blue = Band 1.
 - Apply → OK.
 - Export: Right-click layer → Export → Save As → Rendered Image → GeoTIFF format.
5. False Color Composite
 - Right-click the virtual raster → Properties → Symbology.
 - Assign Red = Band 4, Green = Band 3, Blue = Band 2.
 - Apply → OK.
 - Export the result as GeoTIFF.

Results:

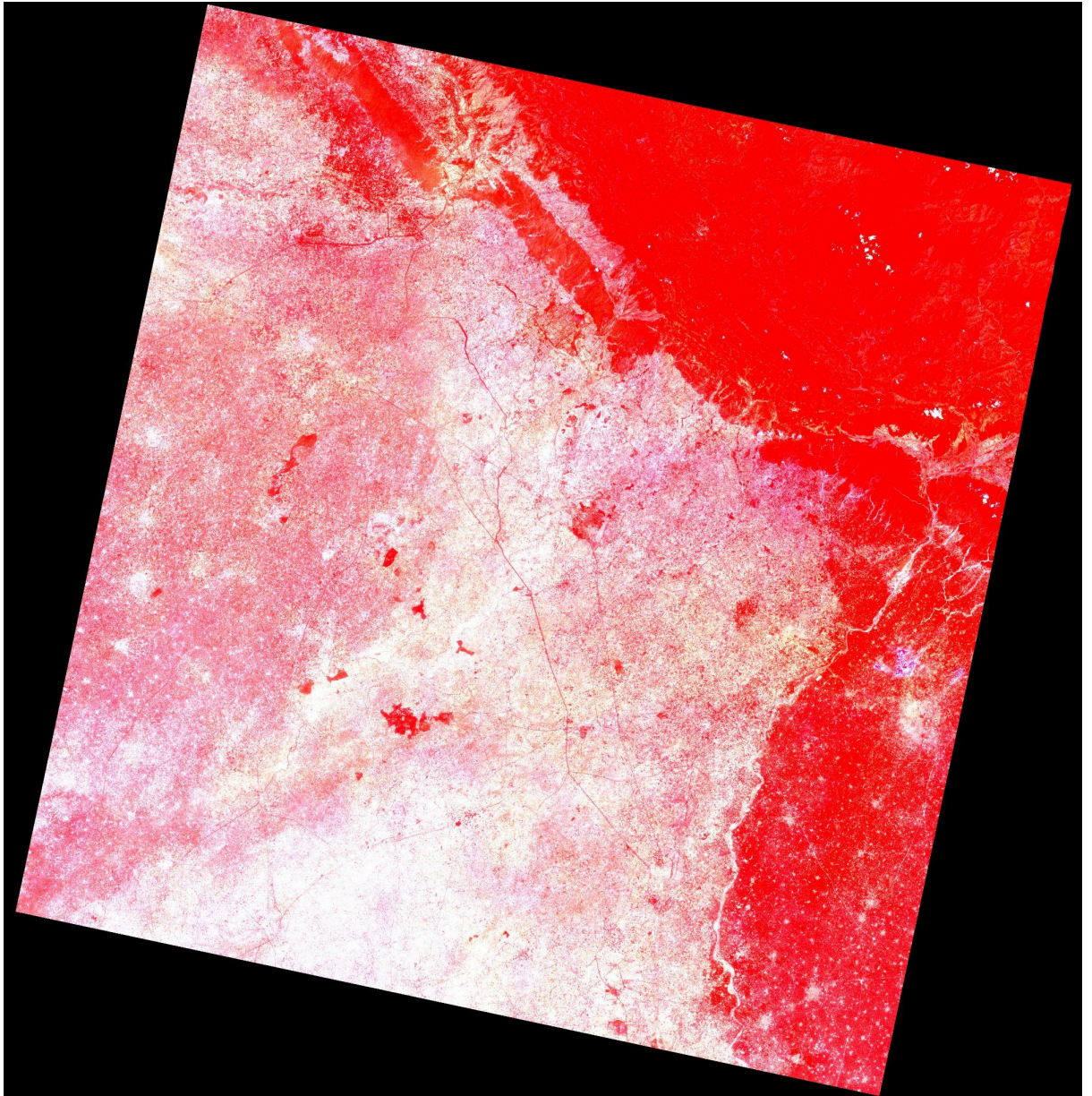
1. Generated True Color Composite (B3, B2, B1) resembling natural colors as seen by the human eye.

Image 1



2. Generated False Color Composite (B4, B3, B2) which highlights vegetation in red and enhances land cover features.

Image 2



Conclusions:

The exercise demonstrated the process of downloading satellite imagery from USGS and creating color composites using QGIS.

- True Color Composite gives a realistic view of Earth's surface.
- False Color Composite is useful for vegetation analysis, land use classification, and environmental studies.

Submitted by

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